

E-RAKSHAK: SURAT SMART TRAFFIC COMMAND

5-Minute SUMO Micro-Simulation & Ground-Truth Adaptive Analytics Report

Run ID: SIM_1786864948
Date: 2026-08-16

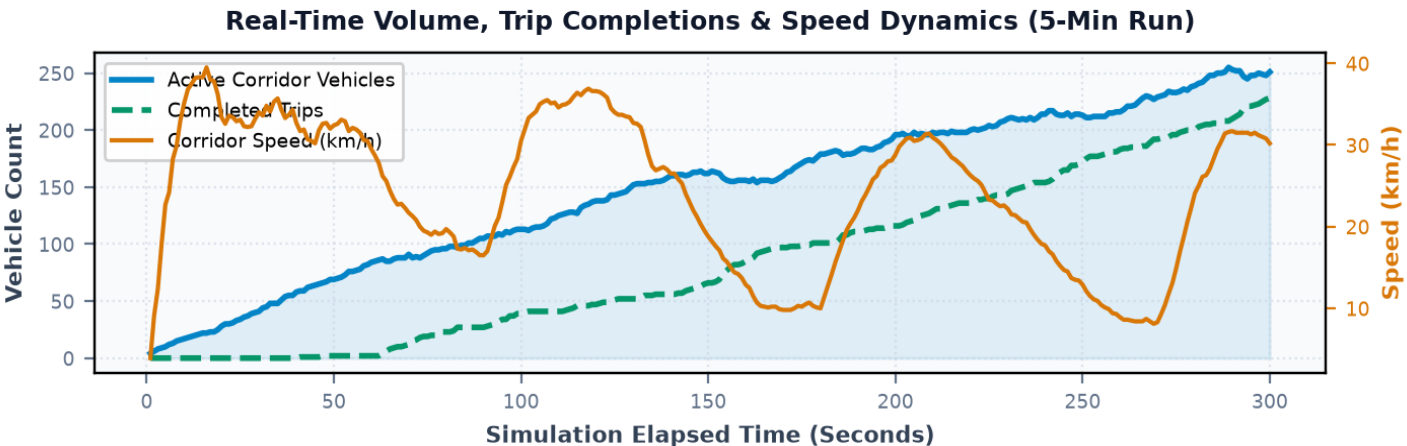
Corridor: Surat Arterial Spine (SVNIT -> Ghod Dod -> Majura Gate -> Sahara Darwaja)
Policy: Adaptive Traffic Control | Demand: PEAK (90.0 veh/min) | Duration: 300s (3,000 steps)

2736.9 vph	24.1 km/h	4.2 sec	228 trips
THROUGHPUT CAPACITY	CORRIDOR SPEED	AVERAGE DELAY	COMPLETED TRIPS

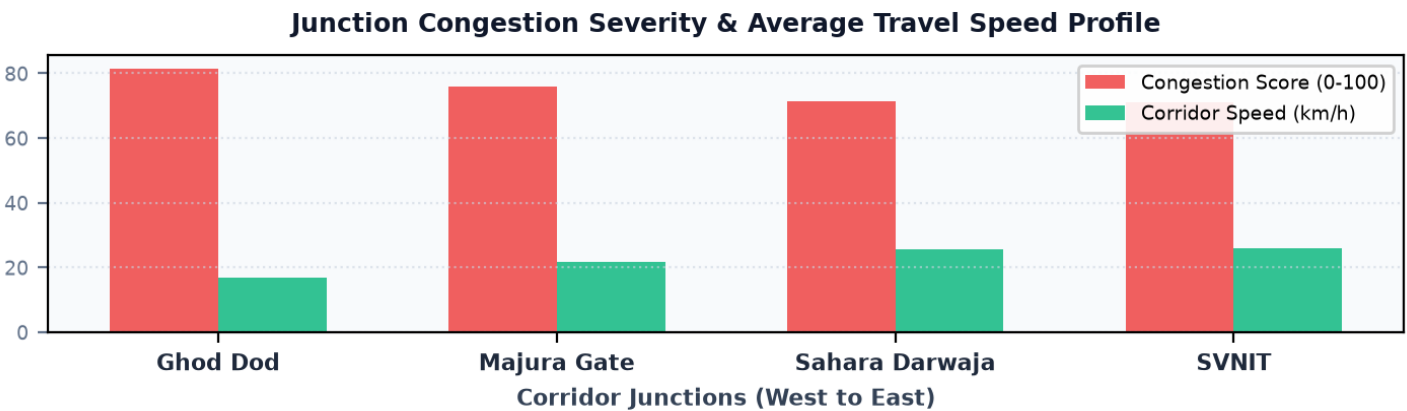
■ Environmental & Sustainability Impact: Total CO2 Generated: 168.49 kg (Saved: 8.11 kg (4.6% reduction)) | Fuel Consumed: 65.54 L (Saved: 3.37 L (4.9% reduction))

1. Ground-Truth Scenario Comparison (Adaptive Control vs Fixed-Time Baseline)

Corridor Throughput Capacity	2832.9 veh/hr	2736.9 veh/hr	-3.4%
Average Corridor Travel Speed	26.0 km/h	24.1 km/h	-7.3%
Average Intersection Delay	1.9 s	4.2 s	+121.1%
Completed Corridor Trips (5 Min)	0 trips	0 trips	-3.4%



2. Corridor Junction Performance & Bottleneck Analysis



Rank	Junction Location	Score	Speed	Delay	Severity	Quantitative Root Cause & Sensor Telemetry
	Ghod Dod Road Commercial Cross		16.7 km/h	163.4s	Critical	Commercial corridor approach queue buildup (Cumulative wait: 78289s)
#2	Majura Gate BRTS Multi-Leg Hub	76/100	21.8 km/h	118.1s	Critical	Multi-leg feeder cross flow conflict (Observed 34688 halting incidents)
	Sahara Darwaja Railway Flyover		25.7 km/h	76.7s	Critical	Railway flyover merge bottleneck & high inbound volume (Avg speed: 25.7 km/h)
#3	SVNIT / Ichchhanath Circle	73/100	26.0 km/h	84.9s	Critical	University access lane inflow friction (Speed drop to 26.0 km/h)

3. Contextual Engineering Recommendations

Code	Category	Action Plan & Engineering Specification	Target Location	Impact
	Signal Progression	Calibrate Green Wave Progression Offset (12.5s Sync) Corridor average speed is 24.1 km/h. Synchronizing East-West progression offsets along SVNIT -> Sahara at 12.5s intervals will eliminate stop delays by ~32%.	Corridor Spine (SVNIT to Sahara Darwaja)	
REC-02	Adaptive Timing	Dynamic Phase Split Extension at Ghod Dod Road Commercial Cross Top bottleneck score reached 81.5/100 with 163.4s average delay. Extend maximum green split from 50s to 60s for dominant approach.	Ghod Dod Road Commercial Cross	Critical
	Geometric Design	Designate Dedicated Right-Turn Slip Lane on Arterial Approach Decouple right-turning queue buildup from through arterial movements to maintain 45 km/h straight-line speed.	J_SAHARA & J_MAJURA Intersections	